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A System for English Vocabulary Acquisition based on Code-Switching

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ABSTRACT

Vocabulary plays an important part in second language learning and there are many existing techniques to facilitate word acquisition. One of these methods is code-switching, or mixing the vocabulary of two languages in one sentence. In this paper the authors propose an experimental system for computer-assisted English vocabulary learning in context using a code-switching based approach for Japanese learners. First they introduce the CO-MIX system, an English vocabulary teaching system that uses code-switching for vocabulary acquisition. Next, they show how they utilize incidental learning techniques with graded readers to increase language proficiency. The authors present the system architecture, underlying technologies, and evaluate the system's performance through user interaction with both a baseline and the proposed system by using a semantic differential scale. They also perform separate factor analysis of participants' attitudes for both systems, an analysis of users' mistakes and compare users' language tests scores. Finally, the authors discuss the evaluation results and further development of this technology.

KEYWORDS

Code-Switching, E-Learning, Intelligent Tutoring Systems, Learning in Context, Natural Language Processing, Second Language Acquisition, Vocabulary Learning

INTRODUCTION

In the modern world, English has become a global language spoken in over sixty countries. English language proficiency is essential for cross-cultural communication and the necessity to know the language is constantly growing. This necessity leads to the problem of finding qualified teachers that can give students a chance to practice their language skills. Vocabulary is the most essential part of second language (L2) learning (Ling, 2010), and recently has become a topic of discussion for researchers, school curriculum designers (Jun Zhang & Annual, 2008; Nation, 2001) and others involved in L2 learning.

The problem of vocabulary acquisition is one of the most significant barriers to mastering foreign languages. In a spoken language, about 1,800 words constitute about 80% of the spoken corpus (McCarthy, 2004). Because some frequent words are often repeated, it is thought that learners can understand a large proportion of foreign language conversation with a relatively small vocabulary

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(McCarten, 2007). The importance of vocabulary is explained by Wilkins, who states: "without grammar very little can be conveyed, without vocabulary nothing can be conveyed" (Wilkins, 1972). However, previous research also suggests that although second language learners need to know a large number of words as a long-term goal, this is not an essential short-term goal (Nation, 2001).

A variety of approaches to vocabulary learning have been presented, but none have been proved to be the most effective way of studying language units. The way that new words are presented to students usually depends on teachers' individual beliefs and teaching strategies. New possibilities for vocabulary acquisition can be found in the code-switching phenomenon, in which a word from one language is used in a sentence in which the grammatical structure belongs to another language. Code-switching presents a chance for students to think about new foreign language words in a deeper manner and offers the potential for expansion of the second language vocabulary. Recently this technique has become a target of interest for researchers in different domains, such as psychologists (Macaro, 2001), linguists (Coady & Huckin, 1997) and computer scientists (Labutov & Lipson, 2014).

Our Contribution

In this paper we propose CO-MIX system, an innovative learning application that can give students a chance to expand their lexical knowledge and utilize the advantages of the code-switching phenomenon. To the authors' best knowledge there exists no robust methodology or teaching application for code-switching based vocabulary acquisition in e-learning. In addition, the code-switching effect on second language vocabulary acquisition has not yet been studied in the field of computer science. The novelty of this research comes from the fact that we present a code-switching based vocabulary acquisition method for the first time in the context of an e-learning system, with evidence to support the effectiveness of our method. We describe the design of the CO-MIX system, and discuss the results of our experiments to evaluate the usefulness of the presented learning application in learning L2 vocabulary.

Another novelty of this contribution is found in combining the phenomenon of code-switching with several other existing methods of teaching vocabulary (see Background section) that have been proved to be effective, and implementing them in e-learning software to facilitate second language word acquisition. An e-learning application should be equipped with a function to encourage users to actively participate in a learning task or increase their motivation to study. Therefore, we utilize the results of our previous study regarding the method of increasing user engagement presented by Mazur et al. (Mazur, Rzepka & Araki, 2011) by introducing various language quizzes to strengthen users' motivation. In this paper we measure users' attitudes towards the proposed system to prove that it is well accepted by the users and properly designed for applying code-switching method in an e-learning system. Moreover, with research results gathered through semantic differential method (described in subsection Research method) we provide evidence suggesting that the users find presented research relevant and useful. Another issue discussed in this work is the problem of automatic creation of study materials for second language teaching. We follow the general idea of an exercise generator presented by Ginsburg (2012) by developing our method for assessing the vocabulary proficiency level of the user with high frequency word lists. The assessment results are used by our system to choose the exercise appropriate to the user's language ability.

Research Goals

The goal of the current research is to present a new vocabulary learning method for e-learning systems that employs code-switching phenomenon. To our knowledge, there has been no research conducted so far on this topic, so investigating how code-switching could be utilized in second language vocabulary

acquisition is a central contribution of this paper. In the current study we address the well-known issue of insufficient vocabulary understanding between second language students in Japan. Thus, new methods of enhancing L2 vocabulary learning are needed to help alleviate this problem.

We also further address the problem of encouraging users' interaction with a system, which has become one of the most important challenges not only in the field of e-learning but in a variety of other domains, e.g., the research in the field of human-computer interaction presented by Dybala et al. (Dybala, Ptaszynski, Maciejewski, Takahashi, Rzepka & Araki, 2010). The idea of a system that would give students a chance to expand their language knowledge and practice with a virtual language partner supporting their endeavor was previously presented by Mazur et al. (Mazur, Rzepka & Araki, 2011).

The main research question we pose in this paper is as follows: "Since there seems to be an agreement on the positive effects of code-switching in learning, can this phenomenon be successfully employed in e-learning to expand students' second language vocabulary?" In order to address the research question we consider two sub-questions: The first one states: "Can the code-switching method enable students to understand words without given definitions?" The latter one focuses on answering how the users respond to presented method and is formulated as: "What are the user attitudes towards our proposed system?"

Our assumptions, based on a preliminary sample of 10 subjects and prior research, include the following: 1. The use of code-switching phenomenon in our system will expand students second language vocabulary; 2. We expect the user attitudes towards our proposed system to be generally positive; 3. We predict the average number of mistakes to be higher for the baseline system; 4. We assume the users will benefit from code-switching method and the scores in language tests while using the proposed system will be higher; 5. Users will be able to successfully obtain the meaning of a number of L2 words from context, without referring to a dictionary.

BACKGROUND

High Frequency Words

In this project we also aim to create an environment in which students can build self-confidence in English communication. One of the ways to achieve this goal is to measure students' second language ability and adjust the difficulty of a given task to their personal level. Our system offers the opportunity for incidental learning using graded readers. These are texts that are graded according to high frequency word count and therefore can be used by learners at different levels of second language understanding. These words are very important because they account for a large proportion of the content in spoken and written texts and are linguistically adjusted to a given user second language level. For example, in the Oxford Bookworm series[1] Level 1 covers 400 words and Level 6 encompasses a total of 2,500 cumulative words. Knowledge of about 4,000 words usually covers 86.0% of a given text (Nation, 2001).

Graded readers are used to support the extensive reading approach to second language learning and provide comprehensible learning materials with word difficulty levels corresponding to users' language ability. The high frequency words of the language are important, and therefore both teachers and learners should spend considerable time on their acquisition. We facilitate vocabulary acquisition through repetition of target high frequency words through a series of tests (quizzes) and presenting comprehensible examples of their usage. If there is a delay between the presentation of a word form and its meaning, students have an opportunity to make an effort to guess or recall its meaning. Nation (2001) states that guessing can only be successful if we provide a good clue to the meaning (incidental learning). Our quizzes offer the chance to experience both incidental vocabulary learning and direct teaching - two popular ways of learning high frequency words.

The problem of automatic generation of teaching materials has been approached by Ginsburg (2012) who proposed a learning program to generate editable English lesson materials to speakers of

Japanese. The author underlines that higher vocabulary skills are crucial in order to understand the words while using this application. He also provides reading exercises with highlighted lexical units and their dictionary definitions. The author underlines that higher vocabulary skills are crucial in order to understand the words while using this application. He also provides reading exercises with highlighted lexical units and their dictionary definitions.

Code-Switching

Code-switching is a phenomenon that exists in many multilingual societies where people use more than one language to communicate on a daily basis, such as Singapore, the Philippines, India, the USA, Spain and China (Velasquez, 1995). It can be defined as "the alternation of two languages within a single discourse, sentence or constituent" (Jamshidi & Navehebrahim, 2013). It can be a brief insertion of a word from one language into the other by two people engaged in a conversation. Other names used to refer to the phenomenon include code mixing and code changing. Code-switching can occur on the lexical level, when there is no existing translation for a word or sentence to L1, or on the semantic level, when an idea can be better explained in second language. There are also two possible locations where the switch can occur, i.e. intersentential (after the sentence) and intrasentential (within the sentence) (Zirker, 2007).

Examples of switching between English and Japanese sentences are given below:

- "*Watashi ha APPLE wo tabetai desu.*[2]" ('I'd like to eat an APPLE.') - English word within a Japanese sentence
- "*She bought it for OMIYAGE.*" ('She bought it for A SOUVENIR.') - Japanese word within an English sentence

The majority of people using code-switching are bilinguals who are able to speak multiple languages. They can easily code-switch and use the method as a resource to find better ways to convey different meanings. Velasquez (1995) states that for bilinguals, code-switching is a linguistic option, because they are proficient in both languages. However, there is an increasing amount of evidence that code-switching can also be used by less proficient people to fill linguistic gaps whenever the learner of a second language encounters the problem of insufficient vocabulary (Nishimura, 1995; Oxford & Crookall, 1990). These two aspects create a common distinction between code-switching as an asset for bilingual people with a high competence in both languages, and as a "reparation tool for insufficiency in the second language" (Hamers & Blanc, 2000). A similar notion comes from Song et al. (Song & Andrews, 2009) who describe code-switching as "an attempt to keep the conversation flowing without having to pause or abandon the message".

In the last two decades of research, there seems to be an agreement on the positive effects of code-switching in learning and that it could be applied in second language learning to facilitate the learning outcome (Celik, 2003; Jamshidi & Navehebrahim, 2013; Song & Andrews, 2009; Lin, 2013). There is a common and proven belief that when code-switching is planned ahead of use, it can greatly contribute to more efficient understanding of a given topic, and be used as a part of second language learning. These effects of code-switching have been studied by Cook (2001) who also found that the use of code-switching in L2 studying does not have a negative effect on learning a foreign language, and moreover, that vocabulary learning can be facilitated by mixing two languages.

Another interesting finding is demonstrated in the work presented by Lin (2013), who discovered that code-switching seems to increase the amount of cognitive processing by students. According to Lin, a larger cognitive effort is necessary to process new vocabulary, which means that students will learn new vocabulary more thoroughly. Although Lin's research did not show conclusive proof for higher effectiveness of code-switching for learning new vocabulary, it indicated that code-switching does not affect the vocabulary acquisition in a negative way.

In our research we focus on intrasentential code-switching, where switching takes place inside a sentence. We also incorporate an idea introduced by Celik (2003), who presented the assumption that in the human mind second language vocabulary is organized in the same way as synonymic expressions are organized in the first language (L1). Celik's research discusses that a link potentially exists between the first and the second language by means of associations and semantics. This link would allow interlocutors to consciously switch elements between two languages. Celik also presents some benefits of code-switching technique for more successful learning of new vocabulary in a storytelling context.

Some attempts have been made to implement bilingual teaching based on code-switching in an online e-learning system, like E-Learning System for Code-Switching Bilingual Teaching by Shi et al. (Shi, Liu & Chen, 2013). However, this Google Translate-based method and the general focus on improving students' international employment competence are completely different from our approach. Their research also lacks any reliable experimental data and does not seem to contribute significantly to the field of e-learning systems and code-switching based methods. Abdali (2010) notes that code-switching does not negatively affect the acquisition of new vocabulary except for minor spelling problems, and that even a simple text can be sufficient to present the new vocabulary to students. The author mentions that some problems with proper spelling of acquired words may occur, but the possibility of wider application of code-switching as a teaching device was suggested.

Incidental Learning and Extensive Reading

A study by Nation (2001) shows that a considerate amount of second language word acquisition happens through incidental learning. The author also stresses the fact that the long-term vocabulary growth is greater via incidental learning. The outcome of this research is the incidental learning theory, inspired by the way children learn vocabulary in their first language. The theory states that the vast majority of vocabulary is learned gradually through multiple exposures to a word over time. However, a typical student who tries to learn a second language vocabulary will encounter the widespread tradition of studying through rote memorization and making word lists. This vocabulary learning strategy comes from the previous century and offers only a minimal amount of context, and therefore is widely criticized and not generally recommended by teachers. For example, an article by Oxford et al. (Oxford & Crookall, 1990) expresses concern that students might not be able to use new word pairs in any communicative way without further assistance.

Another idea of how to improve vocabulary memorization that contributed to our research originated in a study by Nagy et al. (Nagy & Herman, 1985). This study claims that extensive reading should be promoted, because it can lead to greater vocabulary growth. Further research by Nagy et al. (Nagy, Anderson & Herman, 1987) also showed that even one or two exposures to a target word are able to contribute to a studentsould be promotedlary knowledge. Krashen believes that language learners can learn new vocabulary and spelling more efficiently by receiving comprehensible input while they read, and postulated this concept in the Input Hypothesis (Krashen, 1985).

This was also confirmed by Day et al. (Day, Omura & Hiramatsu, 1991), who carried out a similar study with Japanese high school and undergraduate university students. The experiment subjects read an adapted version of an English story and then took a vocabulary test on some of the words from the story. The subjects who read the story learned significantly more words than the control group who only took the test. This outcome provides some insights into the way we study vocabulary and proves that extensive reading has a highly contributive role in the process. Horst et al. (Horst, Cobb & Meara, 1998), in a study using a graded reader with two kinds of vocabulary test, found that an average of five new words were learned. This high acquisition came from the effect of the length of the text and the use of a simplified reader where the unknown words do not appear too densely.

In a study by Labutov et al. (Labutov & Lipson, 2014), the authors claim that humans tend to be good at inferring meaning from context and that single words can be understood from the context of the surrounding text in the paragraph. Further proof for the potential effectiveness of code-switching

for language acquisition comes from the recent findings of Borovsky et al. (Borovsky, Elman & Kutas, 2012). This research contributes evidence that even a single exposure to a novel word in a constrained context results in the integration of the word to the readerext and that single words can be understood from the context of the surrounding text in the paragraph. Further pry high frequency, and shows that words can be remembered incidentally while reading. Coady et al. (Coady & Huckin, 1997) have stated that many adult L2 learners are acquiring new words through reading to improve language proficiency. It has been emphasized that repeated exposure to new vocabulary is needed for a progressive learning process, however previous studies in both first and second language acquisition by Horst et al. (Horst, Cobb & Meara, 1998) did not specify how much exposure is necessary for effective vocabulary acquisition.

Teaching Vocabulary

There are many existing methods of teaching second language vocabulary. One of these is the "diglot method" presented by Christensen et al. (Christensen, Merrill & Yanchar, 2007). In this approach foreign language units are introduced gradually by using a first language storybook. This method requires a human translator who manually prepares the book. The reader is exposed to different words and phrases expressed in the foreign language, but the shortcoming of this method is that student has to look up new words in a dictionary to understand the contents of the book.

Teaching vocabulary mostly involves making students remember new language units by presenting words many times before students can successfully memorize them. Repetition is one of the popular techniques, along with actively recalling a word to be learned. It is also important to organize words in meaningful ways to make it easier to learn, as is often presented in textbooks, as thematic sets benefit the memory.

The other popular way to teach vocabulary is personalization, e.g., using the vocabulary meaningfully to form statements about the student's everyday life, as presented in the work by Schmitt (2000). It is common to encourage students to make their own vocabulary notebooks which consist mostly of written translations of single words in a list, but this method seems to be very limited without actual sentences and dialogs that represent the meaning of words (McCarten, 2007).

Coady et al. (Coady & Hucking, 1997) claim that acquiring a substantial vocabulary in the second language is necessary to attain competencies in other language skills, such as listening, speaking or writing. This research follows Krashen's belief that "language learners acquire vocabulary and spelling more efficiently by receiving comprehensive input while reading" (Krashen, 1985). We follow this general idea with our graded readers approach. This assumption is in strong opposition to the conventional rote memorization technique of lexical studying. Therefore, it can be asserted that new vocabulary units should not be presented only with their first language equivalents and without the presence of verbal second language context.

Gibbons's research (Gibbons, 1987) indicates that nouns are the most often code-switched word class. As many single lexical item switches are nouns, in our study we decided to utilize this finding in order to make the best use of the code-switching phenomenon. Further evidence of the importance of nouns is presented in the studies by Lanza (1997) and Vihman (1998), who also proposed that nouns are more commonly used for the embedded language, i.e. a given base language with the inserted elements of the second language, because translations to the second language are often provided for nouns. A research paper by Yen (2009) also contributes evidence that English often acts as the embedded language with the base Mandarin and Cantonese languages in Hong Kong, and that nouns are the most frequently mixed part of speech.

CO-MIX SYSTEM

System Introduction

CO-MIX is an English vocabulary teaching system that uses code-switching for vocabulary acquisition. The purpose of the system is to introduce a new vocabulary learning method for e-learning systems using intrasentential code-switching, incidental learning and the benefits that are generated with learning vocabulary in context. The following is an example of a Japanese sentence with intrasentential code-switching: "*Migi ni magaru to, konbini ga arimasu. LEFT ni magaru to kouen no geeto ga miemasu*" (If you turn right, there is a convenience store. If you turn LEFT, you will see a gate to the park").

The modular architecture allows easy extensions of the system with additional functionalities. Our system currently has two modules. The first one is responsible for the automatic generation of study materials based on given graded readers in L1 (Figure 1). Its functions are described below. The other module of the CO-MIX system, the Language Quiz Generator, checks the users' understanding of L2 vocabulary through a series of language tests. These quizzes use different strategies to prompt students to match second language vocabulary with their counterparts in the L1 (Figure 2). The system also responds to users' mistakes by providing them with example sentences that show the correct usage of a given word.

In the three available test phases we use different language quiz strategies, e.g. spelling exercises to reinforce students' understanding, or presentation of phonetically similar words to make students'

Figure 1. Process of study material generator

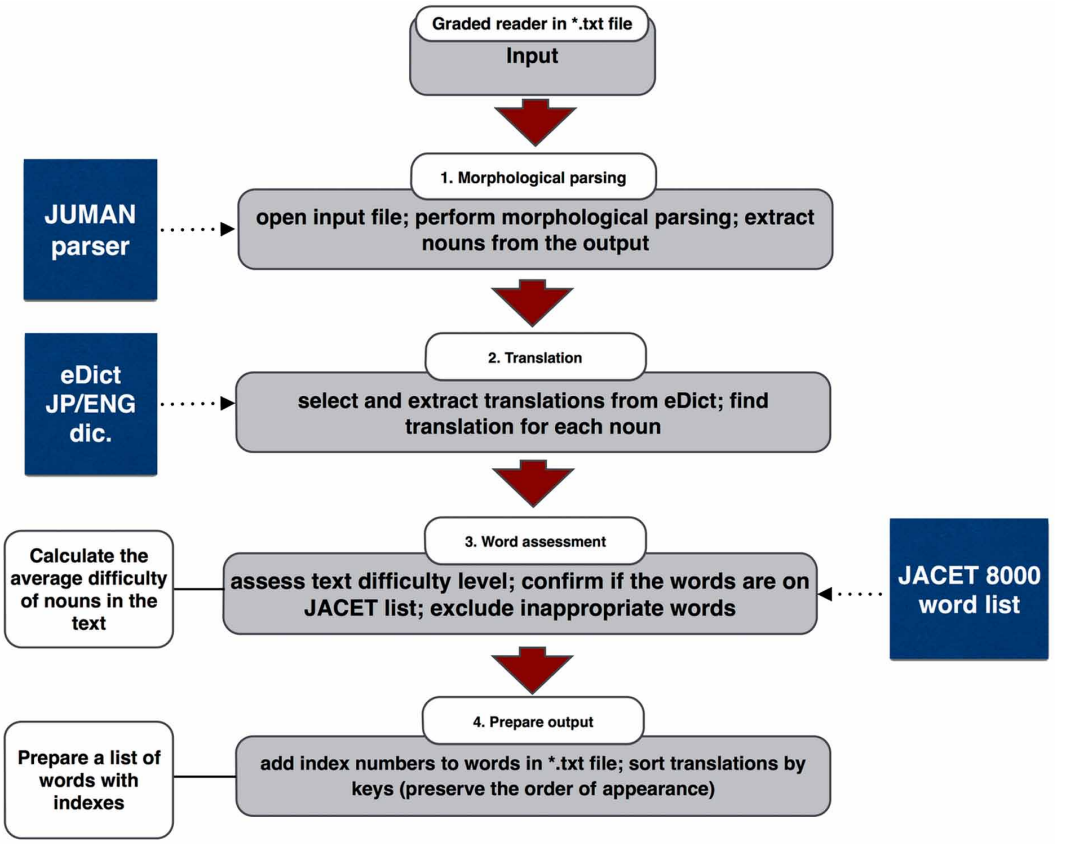
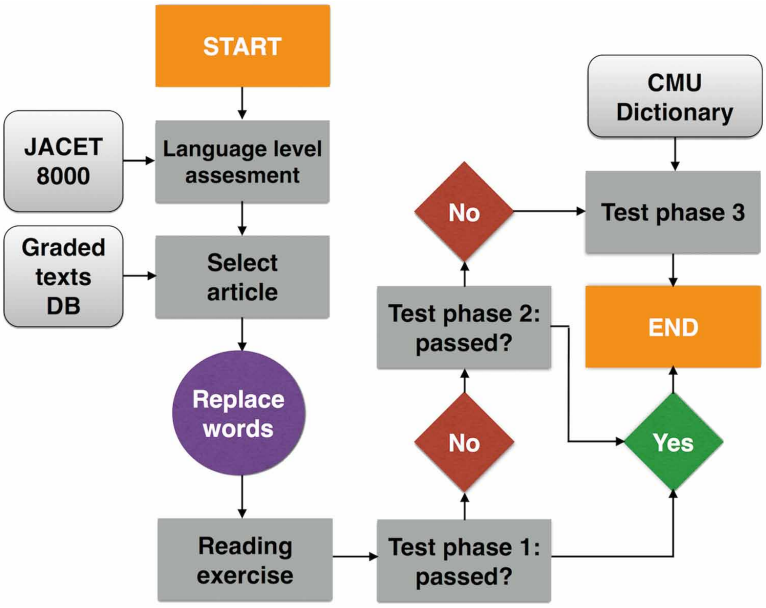


Figure 2. Flow process chart of the language quiz generator



aware of pronunciation similarities between English words (Figure 3). Since each user possesses a different level of English vocabulary knowledge, in the first step the CO-MIX system evaluates user language competency and presents a study article appropriate to the user's English vocabulary knowledge level. Graded readers are used to introduce new vocabulary units and are followed by the vocabulary understanding test phase (Figure 2). The current version of the system is provided with a selection of readings (graded articles) in Japanese for the proposed system, and their translations into English used by the baseline system.

Systems Used

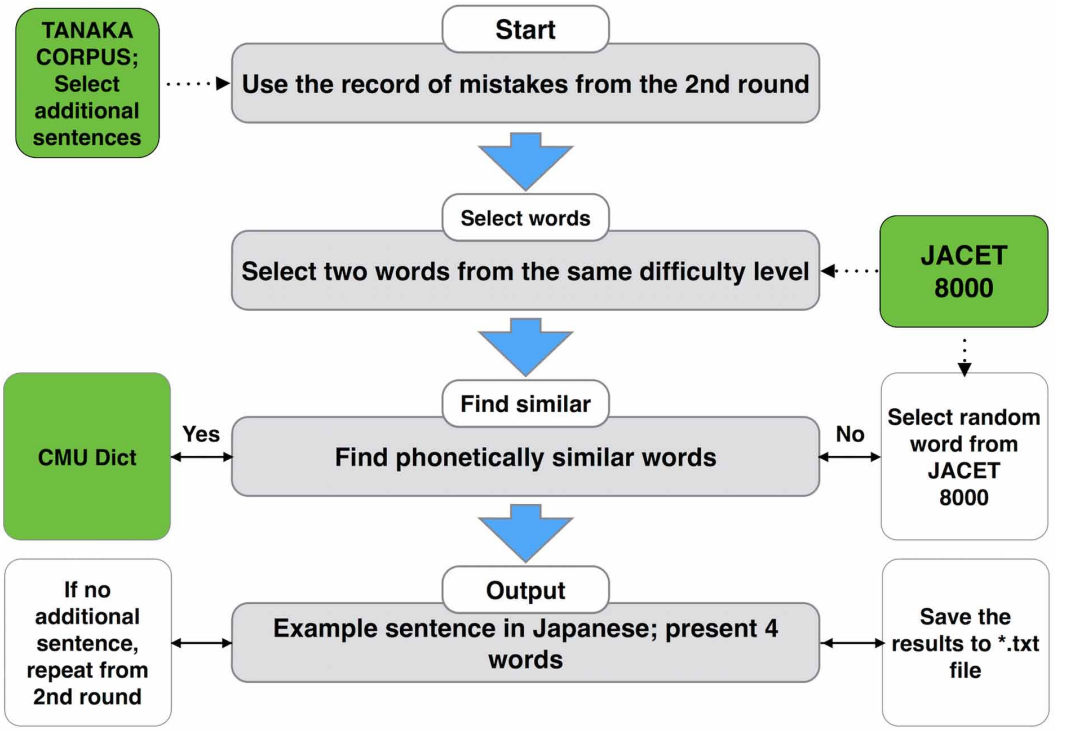
In this research we conducted comparative evaluation experiments using two systems. The baseline system is a variation of our proposed system without the code-switching module. It is equipped with some commonly-used vocabulary learning techniques described above, such as repetition and the diglot method; however, it lacks the advantages of the intrasentential code-switching method presented in the proposed system. Another difference comes from the fact that instead of using readings in a mixed language with intrasentential code-switching, the baseline system presents reading exercises entirely in English using the same set of articles manually translated and proofread by a native speaker.

Study Material Generator

The main function of this application is to automatically create editable learning materials from a given text file (selected graded readers). The study material generator is fed with text files with graded readers in English and Japanese and automatically generates an editable output file with tagged vocabulary items (nouns), their translations in English for the proposed system, and their positions on the JACET 8000 vocabulary list (Uemura & Ishikawa, 2004). JACET 8000 covers approximately 85% of the words used in the graded readers and provides the average difficulty of a text, which allows the system to categorize the text to an appropriate difficulty level.

Usually, preparing suitable readings for students presents a challenge for teachers. The study material generator can be used to manually create additional learning materials from given input

Figure 3. Process of third phase of the test



texts. Selected materials must have a high percentage of vocabulary appropriate to the specific group of learners in order to help students to properly infer word meanings. JACET 8000 is a list of basic words designed specifically for English learners in Japan. It is suitable for our system's needs because words are presented in different levels of difficulty in accordance with their frequency and educational significance. The list is considered to be the most reliable and usable English high frequency word list for graded readers in Japan, mostly for high text cover rate including the vocabulary in both easy reading materials and contemporary English articles (Uemura & Ishikawa, 2004).

The process of the study material generator is as follows. The first program searches the input file for nouns and checks if their English translations are available in the dictionary. Then the program analyzes the word rankings (difficulty level) in a given reading and an average difficulty level is provided for the whole article. In the next step, the system generates an output text with selected words preceded by an index number and supplemented with a list of all chosen words, their proposed translations and their difficulty ranking. In this step, we allow the human teacher to review the result and eliminate possible mistakes that may be caused by limitations of the JACET 8000 list word coverage. This is a necessary step before handing over the materials to the language quiz generator (explained in the subsection below) and presenting them to users. A few search rules had to be established.

In this study we focused on words that are nouns, have an available translation in the dictionary, have a translation that consists of no more than three words, are on our frequency list and are not on the list of the first 500 most common words. This list includes some frequent words that students are most likely to know. The last condition was introduced to exclude words like: "the", "is", "at", "which", "on", etc., since the majority of students will know these words anyway.

Language Quiz Generator

The language quiz generator utilizes reading materials prepared by the study material generator. In the first step, it checks users' vocabulary level and provides them with reading exercises corresponding to their language level. It also generates quizzes to check the understanding of given vocabulary units after completing the selected reading exercise.

The program works in the following manner: the user is presented with a set of 20 polar (yes/no) questions to determine the initial level of language proficiency. Next, the user is assigned to one of the five levels of English proficiency according to his/her final test score. We used the JLPT (Japanese Language Proficiency Test) scale[3], which corresponds with the JACET 8000 vocabulary list. Based on this selection, an appropriate text is selected and presented to the user in Japanese with randomly selected nouns replaced with English words. After finishing the reading exercise, users confirm their readiness to start the test phase. Next, users are presented with a set of a maximum of 20 test questions. The number of test questions depends on the user's language level, the difficulty of the selected reading and number of nouns extracted from the input file.

The main purpose of these quizzes is to bring new vocabulary into productive use and enhance its memorization. The objective of each question is to type the word in English in response to the given vocabulary unit in Japanese. In the first phase of questions, random letters (not more than three and depending on word length) are presented to the user. If the user provides the system with correct answers, the test ends and automatic grading is provided. In the case of error, the system searches for an example sentence with the proper use of a given word and presents it to the user. We use sentences from the Tanaka corpus[4], a large public-domain collection of parallel Japanese and English sentence pairs[5].

Next, mistaken words are repeated in the second phase of questions, where the difficulty level is slightly higher due to lack of support from random letters and the necessity to write the target vocabulary in the sentence. Similarly to the second phase, the system repeats user mistakes in the third and final phase, where four words are presented and user has to input the correct one.

However, one of the randomly selected words out of three is presented with phonetically similar variation, to make students aware of pronunciation similarities between English words and to indicate some vocabulary that could be easily mistaken, e.g. write: reit, rite, wright; tale: tail; nose: noes, noos. We used the CMU Pronouncing Dictionary, a machine-readable pronunciation dictionary, to select words with similar readings[6]. To extract target vocabulary items from texts, we used JUMAN Morphological Analyzer for Japanese[7]. Edict dictionary (Japanese-English)[8] is used to extract English translations of word units.

EVALUATION EXPERIMENTS

Participants

Forty test subjects (university students and adults) participated in two evaluation experiments and interacted with the proposed and baseline systems. The participants were recruited through flyers posted at the local universities and community centers. Twenty-four participants were university students (undergraduate and graduate) and 16 were adult English learners (aged between 24 and 43) who volunteered to participate in the survey. The average age of the participants was 28.3 years. Among the respondents 21 were male and 19 were female. The participants were selected using the following criteria: due to the fact that presented method requires basic English skills, such as reading ability and basic comprehension of English grammar, the target group were people who had some background in English education. Therefore, we performed experiments with Japanese university students and adults who had completed mandatory English education, thus represented comparable language ability. We only accepted test samples if the number of years the participants studied English was similar. Thirty-eight respondents started English education in junior high school, and only two

had been studying English from elementary school. Among the adults, eight people (50.0%) had been studying English for less than two years after graduating, three people for two to five years, and five people for more than five years.

Procedure

The procedure of the experiment was as follows:

1. Respondents were asked to interact with two presented systems. Sessions were conducted in a computer laboratory with 4 identical personal computers. Up to 4 respondents were tested simultaneously. Respondents were seated separately. They were informed about the basic research objectives. Although the delivery of instructions was fully automated, the experimenter remained present to answer possible questions and to prevent interaction among respondents.
2. Respondents were asked to fill out a paper survey after finishing the learning tasks with both systems. All participants were informed about the purpose of the study and the confidentiality of the results. Respondents firstly filled out basic information: age, gender, and English education background.
3. Respondents rated both systems by marking 14 word pairs on a 7-point scale. The same test subjects participated in both experiments, but they were assigned to either start the experiment with the proposed or baseline system; i.e. half of the participants first interacted with the baseline system to assure the reliability of the results.
4. Respondents ranked both systems based on their overall performance.

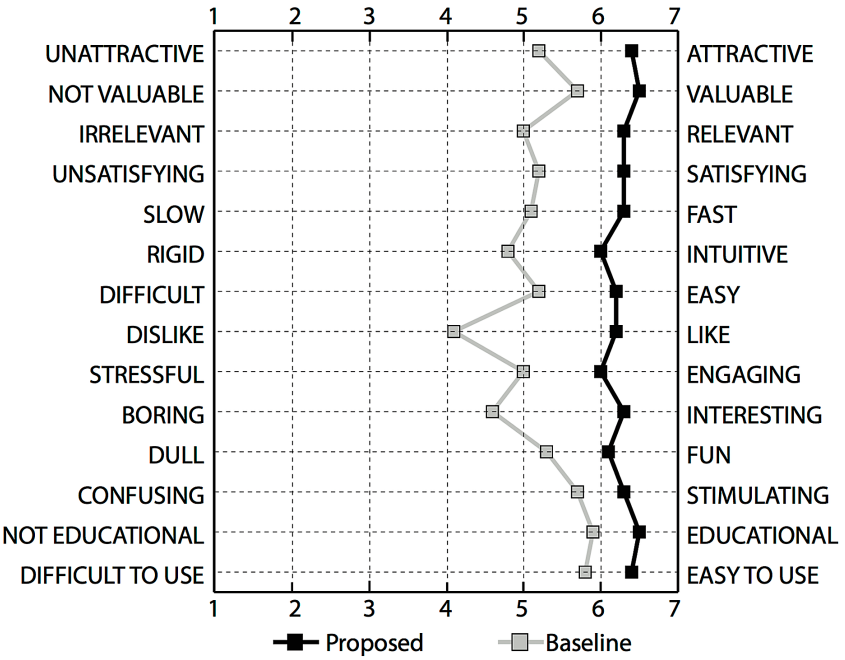
Experiment Method

In order to investigate users' attitudes towards the proposed and baseline systems, we used the semantic differential (SD) scale created by Osgood et al. (Osgood, Suci, Percy & Tannenbaum 1957). Since it has been introduced, the SD Scale has been used in a variety of studies and number of contexts to measure people's attitude towards stimulus words and concepts. Heise (1965) describes SD scale as simple, effective mean for obtaining data on people's reaction towards a given object and provides many evidence that support the validity of the SD as a technique for attitude measurement. In recent research, Urabe et al. (Urabe, Rzepka & Araki, 2014) used the SD Scale to investigate users' perception of an emoticon recommendation system; Ardito et al. (Ardito, Costabile, De Angeli & Lanzilotti, 2006) proposed semantic differential method to analyze users' satisfaction and usability in evaluation of e-learning systems.

The SD scale utilizes polar word pairs such as ovides many evidence that support the validity of thworthless"orthlessale utilizes polar woriven concept. The position marked 0 is labeled "neutral," the 1 position is labeled "slightly," the 2 position "quite," and the third position "extremely. " Ratings on SD scale are often correlated and the average score for each pair of bipolar words indicates the correlation between the word and the object being tested. According to Heise (1970), three major dimensions of word meanings usually account for most of co-variation in ratings. Heise (1970) gives an example of numerous studies that used the SD scale and discusses the three major dimensions of words meanings: Evaluation, Potency and Activity (EPA). They were isolated through the factor analysis by analyzing hundreds of word pairs related to these three dimensions.

In our experiment we decided to use 14 bipolar words (shown in Figure 5) and a seven-point scale to derive subjects' attitudes towards the given concepts and measure their opinions on a metrically controlled scale. Additionally, the polar word pairs were drawn from three general dimensions of word meanings: Evaluation (e.g., valuable – not valuable), Potency (e.g., attractive – unattractive), and Activity (e.g., fast – slow). The bipolar words were determined to represent all major dimensions of word meanings in order to evaluate the participants' attitude towards a computer system. In selection of appropriate word pairs we referred to semantic differential profiles suggested by Heise (1965),

Figure 5. Semantic differential scale for proposed and baseline systems



and works by Urabe et al. (Urabe, Rzepka, Araki, 2014), Osgood et al. (Osgood, Suci, Percy & Tannenbaum 1957) and Newby (2010). We also performed randomization of word pairs by presenting concepts in a random order.

Experiment Results

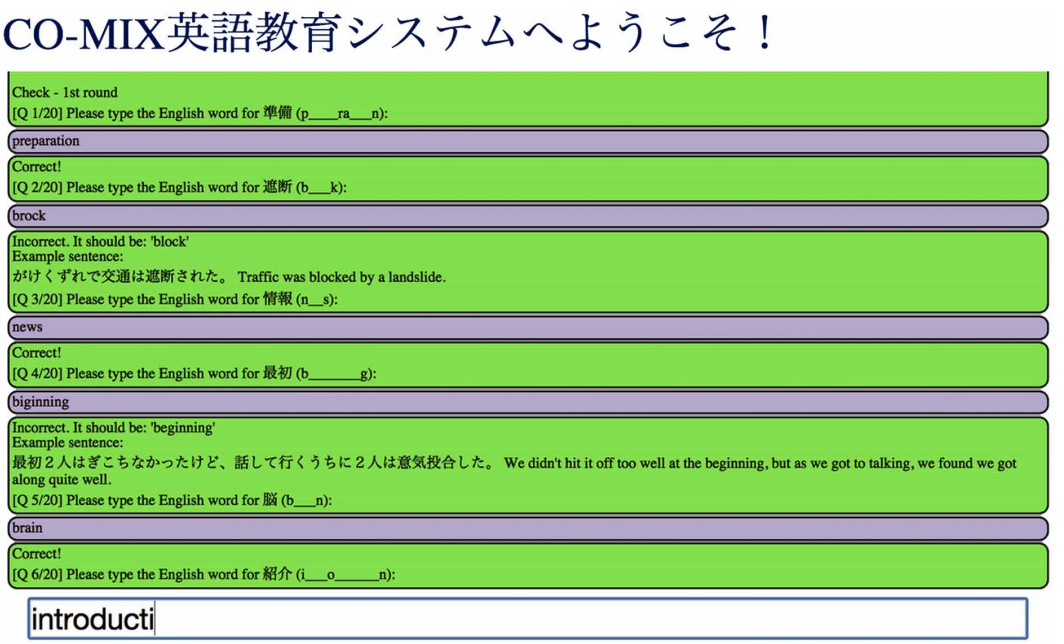
The language level of participants was determined by their number of correct answers in the initial vocabulary test described above (numbers close to zero represent an elementary level of word understanding; numbers close to five represent an upper-intermediate/advanced level of English vocabulary knowledge). The average vocabulary level was 3.61 for the proposed system (eight people reached the maximum score of five, and the lowest score was 0.7). As for the baseline system, the average score was 3.18 (three people finished the vocabulary test with the maximum score, and the lowest result was 0.4).

In the test results for the proposed system, six people reached 100.0% correct answers in the first round (15.0% of the users) for the proposed system, followed by 19 out of 34 in the second round (55.0%) and 12 out of 14 in the third round (85.0%). The average answering time for the proposed system was 11.6 minutes. As for the baseline system, 10 people reached 100.0% correct answers in the first round (25.0%), followed by 11 out of 30 (36.0% of respondents) in the second round and 16 out of 19 (84.0%) reaching the maximum score in the third round. The average answering time for the baseline system was 8.8 minutes.

Participants' Attitudes on the SD Scale

We collected the calculated average of system users' attitudes towards the proposed and baseline systems using the SD Scale. In Figure 4, numbers closer to one represent the part of the spectrum assigned to the words on the left, whereas numbers close to seven are better characterized by the words on the right. The original word pairs in Japanese used in our survey are presented in Appendix B.

Figure 4. Screenshot of system interface



After analyzing the results shown in Figure 5, we reached the conclusion that the average results of the proposed system were distinctly higher than the baseline. The overall average of the proposed system was 6.2 points, which was superior to the baseline system (5.1 points). The general trend line for the proposed system stayed within 6/7 at the higher end of the scale, whereas items in the baseline system such as "like/dislike" and " interesting/boring" diverted from the given trend line. On the other hand, three items ("stimulating/confusing", "educational/not educational", "easy to use/difficult to use") had similar results in both systems, which indicates general user satisfaction with both systems.

Factor Analysis

Two separate sets of statistical tests were carried out on the baseline (System B) and proposed (System A) systems using the result data acquired from the semantic differential analysis (see Figure 5). The reason for this form of analysis was that different questionnaires were given to respondents to evaluate each system independently.

Since putting both sets of data together did not generate reliable comparative results, we decided to perform further analysis, evaluate each set of data separately and then compare their characteristic features. Data from the system evaluations were submitted to Factor Analysis (on SPSS 21[9]). KMO & Bartlett's Test were run on the data for the correlation test and the sampling size to check if the sampling size was fit for factor analysis. The test resulted with 0.91 sampling adequacy for the proposed system and 0.93 for the baseline system, which means that the samples were adequate and therefore fit for factor analysis and the p-value (significance level) for both tests was 0.01.

A second set of analysis, Principal Axis Factoring, was run on the data with the two extracted factors to validate the results of the preliminary statistical test. Items loading <0.50 in whichever factor were deleted for having no statistical significance. For the proposed system, two factors were extracted explaining 55.9% of the data variances. As for the baseline system, 64.4% of the data variances were explained with the extracted two factors.

In the baseline system (see Table 1) the items "Stimulating", "Engaging", "Likable" and "Fun" were extracted with high rates in Factor 1, named Likability. In Factor 2 five items,

Table 1. Factor analysis results for baseline system. The highest significance in bold font

Baseline (System B)	Factor	
	1	2
Stimulating	**0.88**	0.72
Engaging	**0.83**	0.50
Likeable	**0.81**	0.76
Easy to use	0.77	0.60
Intuitive	0.74	0.50
Educational	0.68	0.57
Learning Speed	0.63	
Difficulty	0.57	
Attractiveness	0.55	**0.85**
Fun	**0.80**	**0.85**
Value		**0.82**
User Interest	0.69	**0.80**
Relevance	0.67	**0.80**
Satisfaction	0.60	0.75
Eigenvalue	7.76	1.26
Explained Variance	55.43	9.03
Total	64.46	

Extraction Method: Principal Axis Factoring.
Rotation Method: Promax with Kaiser Normalization.
Items loading <0.50 were deleted for no statistical significance.

"Attractiveness", "Fun", "Value", "User Interest" and "Relevance" named Educational Value were rated highly (see Table 2).

The items for Educational Value were mostly rated highly in Factor 1 for the proposed system, whereas in the baseline system they were extracted in Factor 2, e.g. the item "Valuable" was rated 0.82 in Factor 1 for system A and 0.30 in Factor 1 for system B. It can be interpreted that the item "Valuable" was selected more significantly for the proposed system.

For the proposed system (see Table 2), the items "Value", "Attractiveness", "Stimulating" and "Satisfying" were rated high in Factor 1, named Educational Value. This means that a significant

Table 2. Factor analysis results for proposed system. The highest significance in bold font

Proposed (System A)	Factors	
	1	2
Value	**0.82**	
Attractiveness	**0.80**	
Stimulating	**0.80**	0.67
Satisfaction	**0.80**	0.66
Fun	0.77	**0.70**
Relevance	0.77	0.60
User Interest	0.69	0.51
Learning Speed	0.64	0.57
Easy to use	0.57	
Engaging	0.64	**0.84**
Likeability	0.54	0.68
Educational	0.57	0.65
Intuitive		0.61
Difficulty		0.58
Eigenvalue	6.71	1.10
Explained Variance	47.98	7.92
Total	55.90	

Extraction Method: Principal Axis Factoring.
Rotation Method: Promax with Kaiser Normalization.
Items loading <0.50 were deleted for no statistical significance.

number of the respondents gave high rates to the latter semantic values. "Fun" and "Engaging" were rated high in Factor 2, and named Enjoyment.

Rankings based on the Systems' Performance and Users' Preferences

In the last step we asked the respondents to rank the two systems based on performance, and state which of the two systems they prefer. The results of the survey indicate that the majority of participants selected the proposed system as performing better, and expressed their eagerness to use it in the future. From this result we concluded that our proposed system achieved a significant improvement

over the baseline system in terms of system performance and user preferences. Below we present the results of the ranking:

- **Which system do you think is superior?** 35 out of 40 (87.5%) chose the proposed system
- **Which system would you like to use more after the experiment?** 35 out of 40 (87.5%) chose the proposed system
- **Which system do you like more?** 35 out of 40 (87.5%) chose the proposed system

Analysis of User Mistakes

The average number of mistakes the participants made was higher for the baseline system in all three rounds of the vocabulary knowledge evaluation test (Table 3). The wide gap between the number of mistakes in the first round could support an assumption that users were able to memorize new vocabulary better in the first round. Most of the mistakes the users made can be analyzed by categorizing them in three groups.

The first group, which we labeled as "transfer mistakes", is related to the phonetic matching of loan words and lexical borrowings in the Japanese language. It corresponds to typical phonological problems many Japanese students encounter while learning English. Some examples that fit into this category are mistakes like: "*conbersashon*" or "*conberstation*" (conversation), "*pamito*" (permit), "*pasonal*" (personal) and "*abusensu*" (absence), "*curassu*" (classroom), "*libing*" (living), "*buroku"* (block) and "*elejible*" (eligible). The words in parenthesis are user input words and the correct answers are shown in brackets.

The second group we refer to as "vocabulary mistakes", and they include some obvious examples of erroneous target L2 word associations, i.e., students associated the target English word with some other word they may previously have heard. For example: "*creation*" (criterion), "*convocation*" (conversation), "*action*" (activity), "*thick*" (trick), "*mood*" (mind), "*conference"* (conversation), "*scene*" (screen), "*mails*" (means), "*root*"(rest), "*becoming*" (beginning) and "*preparation*" (presentation).

The third group we distinguish as "spelling mistakes", which contains typical spelling mistakes that happen to many learners of second language: simple performance errors that include random guesses, slips, typos, etc. For example: "*burau*" (bureau), "*univercity*" (university), "*wintin*" (within), "*bycicle*" (bicycle), "*interduction*" (introduction), "*studying abroud*" (studying abroad), "*hourely pay*" (hourly pay), "*expearience*" (experience), "*libing*" (living), "*biginning*" (beginning) and "*screan*" (screen).

DISCUSSION

Experiment results seem to concur with the findings by Chen (1998), who showed that second language learners commonly lack sufficient word knowledge to support proper guessing of the word meaning from a given reading context. Therefore, in the case of the baseline system, the chance of comprehending complex graded readers presented entirely in the second language and effectively acquiring new vocabulary was significantly lower. There is a visible gap between the second round of test between two systems, and higher number of correct answers in the case of the proposed system

Table 3. Average number of mistakes for proposed and baseline systems. Lower number in bold

SYSTEM	1st round mistakes	2nd round mistakes	3rd round mistakes
Proposed	**2.62**	**0.62**	**0.05**
Baseline	3.75	1.10	0.07

may indicate higher effectiveness of code-switching method, not present in the baseline system. This confirms our assumption that the users benefit from code-switching method and its presence will have a positive impact on their test scores.

Another insight comes from the fact that the time users spent with our baseline system was shorter. It also indicates the advantage of code-switching based method used in the proposed system, with higher number of correct answers and lower number of users' mistakes. Analysis of user mistakes confirms another assumption we had made, and demonstrates that the number of spelling mistakes was higher for the baseline system. Smaller number of mistakes indicate that the proposed system helped students to better memorize the spelling of new words in a foreign language. Some opinions contributed by the experiment participants (see Appendix A) also indicate that the proposed system was more encouraging to interact with, while the baseline system seemed to discourage some participants to sufficiently elaborate on a given task, and thus was used for a shorter duration than the proposed system.

The results of our experiments answer our first and second research questions and clearly show that the code-switching phenomenon applied to an e-learning system can expand students' vocabulary and enable them to understand many words without given definitions. These findings are consistent with previous research by Mazur et al. (Mazur, Rzepka, Araki, 2010), in which most of the participants successfully remembered and recreated a series of test sentences in a mixed language, thus proving code-switching based method to be an effective way of acquiring new vocabulary.

Judging from both the results and analyses, it can be asserted that user attitudes to our proposed system were positive and in some categories it significantly outperformed the baseline system, which answers the third research question. Additional rankings, based on the systems' performances and users' preferences, also indicated the proposed system as more favorable. Another insight in the proposed system's educational value and general user satisfaction was provided by the factor analysis results. It demonstrated that the users had more positive impressions towards our proposed system, especially in terms of its educational value and users' preferences.

Although the number of answers ranking the proposed system as preferable was the same for each of the questions, there were some differences in the answers. Only three out of 40 respondents indicated the baseline system as their first choice (7.5%) and two participants chose a different system in each category, which may indicate that they saw the proposed system as superior, but liked the other one more (13.0%). There are many possible explanations, one of them emphasizing the baseline system as more similar to the methods of studying previously encountered by some users. For these users the baseline system seems to be more challenging. For further explanation, we included some relevant examples of user comments after using both systems in Appendix A.

CONCLUSION

In this paper we presented CO-MIX, an experimental system for English vocabulary acquisition in context using a code-switching based approach for Japanese learners. Further experiments that contributed to our previous results demonstrated evidence that the proposed method is useful in vocabulary acquisition. Experiments with the proposed and baseline systems together with the analysis of users' results showed that our method is an effective way of expanding students' second language vocabulary. The high percentage of correct answers, making successful connections between meanings of words in two languages and positive comments from the participants (see Appendix A) suggest that a system for learning English vocabulary in context using a code-switching based approach has the potential to provide Japanese learners with engaging and educational learning activities. The participants improved their vocabulary scores after using the application, which demonstrates the pedagogical benefits of the CO-MIX method. The users of the proposed system were also able to

memorize the new vocabulary better than those who used the baseline system. Higher user satisfaction with our proposed system also provides motivation to improve and refine it in a number of different ways in the future.

FUTURE WORK

There are various ways in which our system could be further improved. In our previous research, we created a foundation for further research on artificial tutors (Mazur et al., 2012). The proposed CO-MIX system is a step towards our general idea of creating a conversational system that would act as a second language tutor and not only transmit knowledge to students, but also offer a variety of other functions including a representation of emotions to make it more human-like and react to the changing emotional states of the users.

In the future we aim to incorporate additional functionalities into the CO-MIX system and advance the project towards our initial idea to provide a chatterbot able to process user queries in mixed languages. Such a CO-MIX system enabled with an artificial tutor could teach users second language vocabulary using the presented code-switching method in a human-like conversation. We are also planning to test our method on sentences with emotions expressed by emoticons. This would allow us to deal with shifting attitudes of users towards the system and respond to the problem of keeping users engaged in learning tasks and motivated to continue using the system in the future.

We are also going to take into consideration suggestions provided by the experiment participants in their comments (see Appendix A) and extend our system to support other languages. We believe that this method can be easily implemented in other languages with a proper morphological analyzer, dedicated study materials (graded readers) and high frequency word lists. Currently, we are working on an inverted vocabulary teaching system. This project is a variation of the system presented above dedicated to aid English-speaking users in studying Japanese vocabulary.

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ENDNOTES

1. **Oxford Bookworm Series of Graded Readers:** https://elt.oup.com/cat/subjects/graded_reading/?cc=global&selLanguage=en_
2. From here onward romanized transliterations of Japanese are italicised. English translations in parentheses accompany the transliterations when needed.
3. "The JLPT is offered in five levels (N1, N2, N3, N4, N5). In order to measure Japanese-language proficiency as thoroughly as possible, test items are designed for each level. N4 and N5 measure understanding of basic Japanese that is mainly learned in the classroom. N1 and N2 measure understanding of Japanese used in a broad range of actual everyday scenes. N3 bridges the gap between N4/N5 and N1/N2."- Japanese Language Proficiency Test, accessed 1 October 2014, < https://www.jlpt.jp/e/about/points.html>.
4. **Tanaka Corpus of parallel Japanese-English sentences:** http://www.edrdg.org/wiki/index.php/Tanaka_Corpus_
5. Tanaka Corpus is an ongoing project and it still contains a number of errors and sentences that may not sound natural in English or Japanese.
6. **The CMU Pronouncing Dictionary: The Carnegie Mellon University Pronouncing Dictionary:** http://svn.code.sf.net/p/cmusphinx/code/trunk/cmudict/_
7. **JUMAN System, a User-Extensible Morphological Analyzer for Japanese:** http://nlp.ist.i.kyoto-u.ac.jp/EN/index.php?JUMAN_
8. **EDICT (Electronic Dictionary);** http://www.edrdg.org/jmdict/edict.html_
9. **IBM SPSS Statistics 21**, one of the leading statistical software used to validate assumptions, analyze and understand data. http://www-01.ibm.com/software/analytics/spss/
10. Some participants provided optional survey comments in both English and Japanese. We translated Japanese comments to English and corrected English comments for minor typos and grammar mistakes.

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APPENDIX A

Selection of Participants' Comments on Performance of Both Systems

Examples showing participants' comments after using the proposed (System A) and baseline (System B) systems[10]:

- "Mixing Japanese and English is easy to understand and interesting, so I enjoyed using system A"
- "I like both but I feel that System A helps me to gradually learn without too much material at once.
- "System B is for someone more confident in his/her language ability"
- "I would like this system to also have a listening exercise, which would make it even better."
- "I had a great time learning with System A. I learned and could remember many new words! I'd like to use it more!"
- "Mixing Japanese and English is easy to understand and interesting, so I enjoyed using System A."
- "I like both but I feel that System A helps me to gradually learn without too much material at once. "
- "Presenting new vocabulary inside Japanese readings was interesting and allowed me to guess their meanings quite well. On the contrary, the texts in System B were entirely in English and I often felt lost and even though the exercise was enjoyable, I don't feel like I have learned many new words."
- "I had a great time learning with System A. I learned and could remember many new words! I'd like to use it more!"
- "System A is better for beginners, and B for people with advanced level of English. A is better for high school students, while B would be better for university level students."
- "It was unique to see English mixed with Japanese language in that specific way. It also made learning new vocabulary easier for me. I think that both systems are interesting and they work pretty well!"
- "I liked the idea of repetition to learn the words which were previously mistaken."
- "System A is better for studying vocabulary and System B for studying grammar."
- "I think it is a very simple and easy way to learn new words. I always learn collocations from sentences. Learning from context helps much more than just making a list of words to remember. I can know the right way to use word in a sentence and I don't like just memorizing words from lists."
- "I liked both systems, but A seemed more effective."
- "It is hard for me to learn vocabulary, because I don't know how to do it well. But, while using System A I could see English vocabulary inside Japanese sentences and understand its meaning. I think it is very good!"
- "If study texts are entirely in English, probably many people will stop reading in the middle. I think that placing new vocabulary inside Japanese texts is a good way to increase the knowledge of words. I just wish there were even more words available. The test was fun, but I would like to see even more exercises, because the mastery of English spelling is a very difficult problem to many Japanese. I also think that schoolgirls would appreciate a cuter interface with some nice looking characters!"
- "I think B is good for reading short sentences and extending writing proficiency in English. In the case of System A, I was able to read it easily and it was a nice and unusual feeling that I can learn English while reading something in Japanese (I think many Japanese often have problems with reading quickly in their own language). Both systems are interesting and I like that they can check mistakes and show the proper usage on the fly. As a learning system, both are enjoyable to use."

APPENDIX B

Fourteen Japanese Bipolar Word Pairs Used in Our Survey

“魅力がある——魅力がない、価値ある——価値のない、適切な——不適切な、満足だ——不満足だ、早く学べる——学ぶに時間がかかる、直間的だ——直間的ではない、簡単——難しい、好き——嫌い、安心——不安、興味深い——つまらない、面白い——面白くない、やる気がわく——やる気がわかない、教育的である——教育的ではない、使いやすい——使いづらい”